

Raúl Quibén Figueroa

AEROSPACE ENGINEER · OPTIMIZATION & DATA-DRIVEN MODELING

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Aerospace engineer completing a PhD (defence expected December 2026) on aircraft design, trajectory optimization and large-scale data analysis for sustainable aviation. Strong in applied optimization, optimal control and Python-based scientific computing, with peer-reviewed publications, a national innovation award, and research experience in Spain, Italy, the US (Stanford) and Japan (Tohoku).

CORE SKILLS

Optimization & modeling: Multidisciplinary design optimization (MDO), optimal control (Pontryagin's principle, direct methods), numerical optimization, simulation

Programming & data: Python (NumPy, SciPy, pandas, OpenMDAO, pyOptSparse, CasADi), large-scale data analysis, MATLAB, Git, HPC, FEM (Abaqus)

Aerospace domain: Aircraft design & performance, hydrogen propulsion, air traffic management, aviation emissions and contrails (CO₂ and non-CO₂)

Professional: Scientific writing, international teamwork, teaching, research project proposals

PROFESSIONAL EXPERIENCE

Predoctoral Researcher · Aircraft Operations Lab, Dept. of Aerospace Engineering, UC3M Jul 2022 – Dec 2026

- Built a multidisciplinary design-optimization framework in Python for hydrogen-powered and hybrid-electric aircraft, coupling design, performance, propulsion, cost and emissions models.
- Formulated and solved large-scale optimal-control problems (indirect and direct 4D methods) to minimize fuel, time and climate impact, including contrails.
- Member of the HYDROGENATING project; contributed to the SESAR ZENITIA proposal. Taught flight-simulator, flight-planning and H₂-aircraft sessions to BSc/MSc students.

Visiting Researcher · Stanford University, Civil & Environmental Engineering May – Jul 2024

- High-fidelity LES of early contrail formation validated against ground-based camera images (presented at EGU 2025).

Research Collaborator · University of Vigo Feb – Jun 2020

- Drone communication via LoRa technology: programming and system testing.

Engineering Intern · AIMEN Technology Centre May – Jul 2019

- Finite-element analysis in Abaqus for structural calculation, certification and project delivery.

EDUCATION

PhD in Aerospace Engineering · Universidad Carlos III de Madrid 2022 – Dec 2026

- Thesis: H₂-fuelled aircraft configuration and trajectory optimization to reduce aviation's environmental impact. Supervisors: M. Soler, R. Cavallaro.

MSc Aeronautical Engineering · Universidad Carlos III de Madrid 2022

- Taught fully in English. Exchange semester at Politecnico di Milano (Erasmus+).

BSc Aerospace Engineering · University of Vigo 2020

- 2nd Best National Graduate (SEDEA 2020); top 10% of class. Thesis at Tohoku University, Japan (JASSO scholarship).

SELECTED AWARDS & HONORS

- Luis Azcárraga Aeronautic Innovation Award 2025** (Fundación ENAIRE, €10,000) for research on hybrid-electric regional aircraft.
- 2nd Best National Graduate in Aerospace Engineering, SEDEA 2020 · Community of Madrid predoctoral contract 2023 (score 8.98).
- JASSO Excellence Scholarship (Japan) · UC3M mobility grant (Stanford) · Xunta de Galicia Academic Excellence Award.

SELECTED PUBLICATIONS

- Feasibility studies on regional aircraft retrofitted with hybrid-electric powertrains, *Aerospace Science and Technology* (2024). **[Awarded]**
- Multidisciplinary design optimization of hydrogen-powered transport aircraft, *Journal of Aircraft* (under review).
- Optimal cruise of hydrogen transport aircraft, *Aerospace Science and Technology* (submitted).
- MOTIVATION: MDO for sustainable aviation, *AIAA AVIATION 2023* · H₂-powered regional aircraft architectures, *AIAA SciTech 2025*.
- Comparing high-fidelity LES of early contrail formation with ground-based images, *EGU General Assembly 2025*.

LANGUAGES

Spanish (native) · Galician (native) · English (C1, Cambridge CAE) · Japanese (A1) · Portuguese & Italian (beginner)